Featured Projects & Experience Portfolio Dossier

Curated Data Structure & Technical Specifications • Dian Raisa Gumilar

I. SCHOLARLY RESEARCH & MACHINE LEARNING PROJECTS

Integrating BERTopic and Heterogeneous Knowledge Graphs for Temporally-Evaluated Research Collaboration Recommendation

AI & MACHINE LEARNING

id: temporal-hkg-bertopic-collaboration-recommendation

summary: An end-to-end scholarly collaboration recommendation framework combining BERTopic neural topic modeling with a Heterogeneous Knowledge Graph (HKG) and FastNode2Vec node embeddings, rigorously validated under a leak-free temporal link prediction protocol on 51,502 Scopus publications.

tags: BERTopic, Heterogeneous Knowledge Graph, Link Prediction, FastNode2Vec, NLP, Temporal Evaluation, Scopus

overview: Constructed an integrated HKG connecting authors, papers, and topics across Indonesian Computer Science publications (2021–2025) to recommend latent interdisciplinary partnerships through fused topological and embedding proximity.

problem: Structural methods overlook researchers with thematic alignment but no shared co-authors, while standard link prediction models rely on random edge splits that introduce severe temporal data leakage and overly optimistic predictive claims.

approach: Discovered 500 cohesive topics using BERTopic; built an HKG with 149,275 nodes and 2,299,184 edges; scored unlinked topic-sharing pairs via hybrid weighting (0.4 embedding cosine similarity, 0.3 Adamic-Adar, 0.2 Jaccard, 0.1 Common Neighbors); audited and corrected temporal evaluation leakage by strictly retraining embeddings on pre-2025 historical subgraphs.

contribution: Designed the HKG schema, implemented the composite scoring algorithm, conducted the architectural audit that isolated and resolved temporal embedding leakage, and ran macro-averaged ranking evaluations across 16,528 test queries.

results: Generated the Top-20 latent collaboration recommendations. Proved that uncorrected evaluations yield an inflated ROC-AUC of ~0.999, whereas leak-free temporal retraining achieves a robust and genuine performance advantage over structural baselines.

metrics: Topic Quality: C_v = 0.7148, Diversity = 0.655, Outlier Rate = 37.21%. Link Prediction: ROC-AUC = 0.801 (95% CI: 0.797–0.806), AP = 0.847 (vs. Structural Baseline ROC-AUC = 0.686). Ranking: NDCG@5 = 0.656, MRR = 0.651, Recall@20 > 0.99.

Collab Identifier: Institutional Research Thematic Alignment Prototype

NLP & DATA SCIENCE

id: collab-identifier-bertopic-prototype

summary: An interactive research intelligence prototype employing independent institutional BERTopic models, Cosine Similarity mapping, and generative LLM labeling to discover direct and complementary collaboration themes from Scopus metadata.

tags: BERTopic, Natural Language Processing, Cosine Similarity, Research Intelligence, Topic Modeling, Python, Generative AI

overview: Developed an institutional analysis tool processing publication abstracts from PRSDI BRIN and UNP Kediri independently to identify shared thematic strengths and unique institutional specializations without cross-corpus semantic interference.

problem: Conventional bibliometric methods (bibliographic coupling, co-authorship networks) are retrospective and depend on existing connections, failing to discover prospective thematic alignment across disparate institutions.

approach: Built modular pipelines for independent topic extraction using Sentence-BERT and HDBSCAN, integrated Mistral API prompts for concise topic label synthesis, visualized similarities via interactive heatmaps and word clouds, and performed expert panel evaluations.

contribution: Implemented data preprocessing and BERTopic modeling scripts, integrated automated LLM labeling workflows, developed analytical UI views, and conducted sensitivity threshold testing.

results: Delivered an interactive analytical tool uncovering viable collaborative opportunities validated by domain experts, demonstrating that computational similarity paired with human validation provides actionable research intelligence.

metrics: Coherence: PRSDI C_v = 0.4265 (vs. LDA 0.3817), UNP Kediri C_v = 0.4147 (vs. LDA 0.3261). Expert Reliability: Label Validation Cronbach's α = 0.932; Recommendation Validation Cronbach's α = 0.962. Expert Evaluation Mean: M = 3.24 to 3.94 / 5.00.

II. APPLIED SYSTEMS, SOFTWARE & IOT PROJECTS

Recipe Information Retrieval System Using BM25

NLP & DATA SCIENCE

id: bm25-recipe-information-retrieval

summary: A specialized culinary document management and search platform leveraging text preprocessing pipelines and the Okapi BM25 ranking algorithm for high-relevance query matching.

tags: Information Retrieval, BM25 Algorithm, NLP Preprocessing, Stemming, Python, Search Optimization

overview: A search platform designed to ingest recipe documents across varied file formats, apply NLP cleaning steps, and index culinary terms for fast and contextually ranked retrieval.

problem: Unstructured recipe text often produces noisy search results and slow retrieval times in traditional keyword databases lacking term-frequency saturation and length normalization.

approach: Implemented text cleaning pipelines (tokenization, stopword removal, language-specific stemming) and integrated Okapi BM25 scoring with file ingestion validation.

contribution: Architected the text processing engine, programmed the BM25 query-document scoring mechanism, and engineered file upload validation handlers.

results: Delivered a stable, responsive search engine achieving sub-200 millisecond response times while maintaining high thematic relevance across test queries.

metrics (Impact & Benefits): Information Access Efficiency: Enabled instant recipe retrieval with query latency under 200 ms; eliminated manual search friction through automated text parsing and standardized term scoring.

IoT-Based Flood Detection and Early Notification System

MOBILE & IOT

id: iot-flood-detection-warning

summary: An embedded early-warning disaster management system combining Arduino Uno, multi-stage water level sensors, and the Blynk cloud platform for real-time mobile push notifications.

tags: IoT, Arduino Uno, Blynk Platform, Embedded C++, Sensor Integration, Disaster Early Warning

overview: An early hazard mitigation system that tracks real-time water elevation shifts, displays visual threshold statuses locally, and broadcasts instant alerts to connected smartphones.

problem: Sudden residential flood inundations frequently lead to property loss and life safety hazards due to the absence of accessible, localized early warning telemetry.

approach: Interfaced an analog water level sensor with an Arduino microcontroller, mapped readings into tri-tier safety boundaries (Safe < 10 cm, Alert 20 cm, Danger ≥ 30 cm), configured local LCD/LED visual cues, and linked telemetry to Blynk mobile channels.

contribution: Engineered hardware breadboard circuitry, wrote embedded threshold detection logic, and established cloud telemetry transmission to mobile devices.

results: Successfully built and deployed an operational hardware prototype that provides dynamic hazard classification and immediate remote alerting.

metrics (Impact & Benefits): Proactive Risk Mitigation: Shortens disaster response time for local residents through automated mobile notifications; eliminates manual monitoring hazards with automated status classifications.

Smart IoT Hydroponic Monitoring and Actuator Control System

MOBILE & IOT

id: iot-smart-hydroponic-monitoring

summary: An automated environmental telemetry and control system integrating water quality sensors (pH, turbidity), temperature, and humidity monitoring with web-based actuator switching.

tags: IoT, Microcontrollers, Web Telemetry, Sensors, Actuator Control, REST API, Environmental Monitoring

overview: An end-to-end automated cultivation management system transmitting multi-sensor water quality metrics to a cloud server and web dashboard for remote actuator intervention.

problem: Hydroponic crops require narrow nutrient and pH ranges; manual chemical testing is labor-intensive, intermittent, and vulnerable to human oversight.

approach: Configured analog and digital sensor arrays on a microcontroller, formatted telemetry payloads into structured JSON streams, and built a real-time web UI with DC pump controls.

contribution: Integrated sensor calibration routines, developed JSON telemetry transmission pipelines, and programmed bidirectional actuator switching controls.

results: Delivered an active monitoring dashboard providing continuous environmental visibility and remote water pump toggling.

metrics (Impact & Benefits): Operational Precision & Automation: Prevents crop stress by continuously monitoring pH and turbidity; eliminates routine manual measuring tasks through remote actuator automation.

Applied Computer Graphics: Seasonal Changes Simulation

SOFTWARE & GRAPHICS

id: pygame-seasonal-transformations

summary: An interactive 2D/3D visual graphics simulation modeling seasonal transitions, precipitation cycles, and solar rotation using mathematical affine transformation matrices in Pygame.

tags: Computer Graphics, Pygame, Python, Affine Transformations, Physics Simulation

overview: A visual programming project demonstrating mathematical object transformations (scaling, rotation, translation) to model seasonal weather phenomena in real-time graphic frames.

problem: Visualizing dynamic atmospheric behaviors and transformation mathematics can be computationally heavy without optimized coordinate transformation routines.

approach: Programmed matrix transformations for scaling, translating, and rotating graphical primitives; orchestrated event loops and particle flows in Python to emulate rain and cloud dynamics.

contribution: Developed object transformation equations, programmed particle motion algorithms for rainfall simulation, and implemented keyboard/event handling in Pygame.

results: Created an educational graphical simulation demonstrating realistic environmental transitions through lightweight procedural rendering.

metrics (Impact & Benefits): Educational Clarity & Performance: Translates abstract geometric transformation matrices into an interactive educational simulation running smoothly on standard hardware without heavy game engines.

Cloud-Integrated Mobile Educational Quiz Application

MOBILE & IOT

id: mobile-educational-quiz-app

summary: A mobile assessment application integrating cloud REST endpoints and Firebase Firestore for dynamic question distribution and state-synchronized evaluation.

tags: Mobile Development, Firebase Firestore, REST API, Cloud Database, Android

overview: A mobile educational tool designed to stream interactive assessments from cloud databases, enabling dynamic updates to question repositories without client app redeployment.

problem: Hardcoded quiz applications require manual app updates to add new learning content, creating high maintenance friction and static user experiences.

approach: Decoupled content storage from the mobile UI by structuring schema collections in Firebase Firestore and consuming question banks via asynchronous REST API calls.

contribution: Designed Firestore document structures, implemented mobile API ingestion routines, and structured score calculation logic.

results: Delivered an interactive mobile application with synchronized question streaming, flexible score tracking, and clean state handling.

metrics (Impact & Benefits): Content Agility & Engagement: Allows educators to deploy new educational content and quiz banks in real time via cloud endpoints without requiring client updates.

III. LEADERSHIP & STUDENT GOVERNANCE EXPERIENCE

Himpunan Mahasiswa Informatika (HMIF) ITENAS

Staff • Internal Audit Unit (Satuan Audit Internal)
July 2024 – December 2024

Conducted procedural oversight and accountability reviews of organizational work programs, verified internal governance compliance, and audited activity documentation across departmental divisions.

Staff • Cadre Development Division (Bidang Pembinaan & Kaderisasi)
September 2023 – June 2024

Facilitated student onboarding programs, structured leadership development curricula for incoming informatics cohorts, and organized collaborative peer mentoring initiatives.